What an Auxiliary Cue Must Carry: LiDAR Body Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Body pose is an intuitively attractive input for pedestrian trajectory forecasting: gait, stance and facing appear to encode intent that a position history does not. We test this on the Waymo Open Motion Dataset with SMPL poses recovered from LiDAR, and find no measurable benefit. The decisive comparison is not pose against no pose—which confounds body information with the capacity of an extra input branch—but correct pose against the same architecture fed *another pedestrian's* pose. On that control the effect is $+0.66\%$ (p=0.52) from scratch, $+0.88\%$ (p=0.18) after full-dataset pretraining, -1.02% (p=0.56) at an intermediate scale and $+0.04\%$ (p=0.93) at $43\times$ more training scenes, while the baseline improves 19% from data alone; at the largest scale the data bound the content effect to $\pm 1.7\%$. A directed search over the integration code accepted one of 22 candidates under a truncated proxy, and the pre-registered eight-seed confirmation that followed returned -0.04% (p=0.92). An oracle control rules out a dead pathway: the same branch, fed the agent's own future instead of its body, improves minADE by 34.8%. We then measure what that pathway converts. Corrupting the oracle with a controlled angular error leaves the gain large at every accuracy we trained: -13.6% even at a median bearing error of 15.1°, which is worse at every percentile than the tracker's box heading the model *already* receives. The decay separates into a precision term with a 3.0° half-life and a term that survives any corruption we applied, and the second is future-side mode selection: it supplies 53% of the endpoint error reduction against 21% at 2 s, and a channel corrupted to 15.1° still closes 92% of the perfect oracle's top-1-versus-best-of-six gap while recovering 39% of its minADE. So the oracle pays for knowing which future occurred, which no observation-time cue supplies at any accuracy. Body facing is instead a noisier copy of the box heading (12.3° against 3.9° on the same bearing), inherited from the trajectory by a snapping step, and worst exactly where it would be non-redundant: 41.2° for the pedestrians stationary at prediction time. Corroborating this, the tracker's box heading estimates the same quantity more accurately (25.2°) and pose does not beat it in any range, occlusion or heading-jitter stratum. Finally, aggregate MPJPE is the wrong acceptance metric for pose in a perception stack: scored per joint against a mean-pose baseline, the head's MPJPE looks *better* than either wrist while it carries a tenth of their skill.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, negative results**

I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes, and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose.

This paper reports that, on the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy—and, more usefully for practitioners, identifies a specific reason that can be checked quantitatively, and cheaply, before a modality is integrated at all.

We make three contributions.

A controlled null. We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch, and it is the comparison a practitioner should demand before adopting any auxiliary modality. The control is null at three training scales; across six pose encodings (6D joint rotations, 3D joint positions, root-orientation-free, agent-frame root, constant-pose and eight frame-invariant descriptors) and two encoder architectures (GRU and cross-attention); under two integration styles, a directed architecture search, and a pre-registered eight-seed confirmation on a held-out split. An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by 34.8% through the same branch—so the null is a property of pose, not of the plumbing.

What the pathway converts, and what it does not. We corrupt that oracle with a controlled angular error to ask what kind of information the branch turns into accuracy. Every accuracy we trained converts, including one uniformly worse than an input the model already has, so the ladder does not set an accuracy a cue must reach; it separates a precision term with a 3.0° half-life from a corruption-invariant term that is future-side mode selection. Body facing supplies neither: it is a noisier copy of the box heading where pedestrians move,

and where they do not—the case velocity cannot cover—it reaches $41.2°$. We also show the same ladder run at 579 scenes reports a clean accuracy threshold that 25k scenes removes, so an underpowered ladder manufactures a floor where there is none.

A measurement caveat. Aggregate MPJPE cannot distinguish a pose estimator that tracks a joint imprecisely from one that ignores the input and emits its prior. Scoring each joint against an always-predict-the-mean-pose baseline, MPJPE is uninformative about which joints are actually recovered ($\rho = -0.11$ across joints, n.s.) and inverts the ordering for the head. Accepting a pose estimator into an AV stack on a single error number risks importing a prior.

We also report, as a cautionary note, that our first attempt at this study was invalidated by an export bug that silently substituted other pedestrians' bodies—a failure any pose pipeline keyed by object identity can reproduce.

II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers; we build on MTR [1], [2], which is representative and strong. Human mesh recovery from LiDAR is maturing [3], with motion priors used to complete sparse observations [4]–[6]. Intent-focused work on pedestrian datasets [7], [8] targets crossing decisions rather than metric trajectories.

Relation to prior positive results. Pose conditioning has reported real gains; our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [9] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with *ground-truth* 3D pose, and—the harder case for us—from 0.40 to 0.36 on JRDB with *estimated* pose. Social-Pose [10] reports about 25% on JTA ($1.20 \rightarrow 0.90$) and on JRDB. Salzmann et al. [11] report $\approx 10\%$ from onboard-estimated keypoints at ranges up to 7 m, specifically for agents with little position history, which is also our best hypothesis.

Le et al. [12] reach the opposite conclusion on JRDB and, in doing so, say the most useful thing about ours: they localise the gain to *lower-body* 3D keypoints (13% ADE, a further 1–4% from derived biomechanical cues). Their cue is our weakest channel. Of the twelve limb joints in Table IV the knees and ankles that carry gait sit at the bottom of the range—SKILL $+0.46$ to $+0.49$, against $+0.63$ and $+0.64$ for the elbows—and what leg motion we do recover is damped: predicted knee and ankle spread is 84–91% of ground truth, where elbows and wrists reach 96–99%. A panoramic camera at a few metres resolves a swinging leg; a few dozen LiDAR returns at 40 m recover its mean position and lose much of its amplitude. So the two results do not conflict—theirs is what this cue is worth when the limbs are resolved, ours what it is worth when they are a few dozen returns and the trajectory already carries the body's direction.

The most informative comparison is within Social-Pose itself. Alongside JTA it evaluates on Pedestrians and Cyclists in Road Traffic, the one vehicle-captured dataset there, and—with *ground-truth* 3D pose—ADE moves from 0.58 to 0.57 m, by 0.01 m or less for all five architectures tested, against the $\approx 25\%$ the same encoder delivers on JTA. The authors attribute this to the shorter horizon and the absence of interacting neighbours; our measurements suggest an additional reason that travels with the sensor rather than the horizon. Either way, the closest published setting to ours already reports approximately our result.

Two properties separate the remaining positive settings from an AV stack, and both follow from the mechanism in Sec. V. *Range and sensor:* JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. *The baseline:* those predictors consume positions, not a tracker's box heading, so body facing is new information there and partly duplicated here. On controls, none reports a shuffled-pose arm—though the confound differs by design. Social-Transmotion masks modalities within one network, so no extra parameters are involved there, whereas a dedicated pose branch pays the capacity cost we measure ($+1.7\%$ minADE)—the same order as the smaller reported gains.

III. Setup

A. Data and a data-integrity failure

We use WOMD [13] with SMPL [14] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [15], which fits per-frame bodies with LiDAR-HMR [3] and completes them using AMASS-trained motion priors [4]–[6]. Poses cover the 11 observed frames; the prediction horizon is 8 s at 10 Hz. The branch is not starved: every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one, so sparsity of pose is not an explanation for anything that follows.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than $90°$, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s.

Our first pass at this study was invalid. The export keyed pose clips by a scene-*local* object identifier, so clips collided across scenes and nearly every training sequence carried another pedestrian's body. A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from $41.5°$ to $11.4°$ and raised the gait–speed correlation from $r = 0.26$ to 0.58. We report it because the failure is silent—training converges, losses look normal—and identity-keyed joins are common in pose pipelines. Every number here is from the corrected export.

B. Arms and the content test

Three arms share one architecture (Fig. 1). *No-pose* receives trajectories and map. *Pose* adds the pedestrian's own SMPL sequence. *Shuffled* receives another pedestrian's sequence, with identical parameter count, optimiser and input dimensionality.

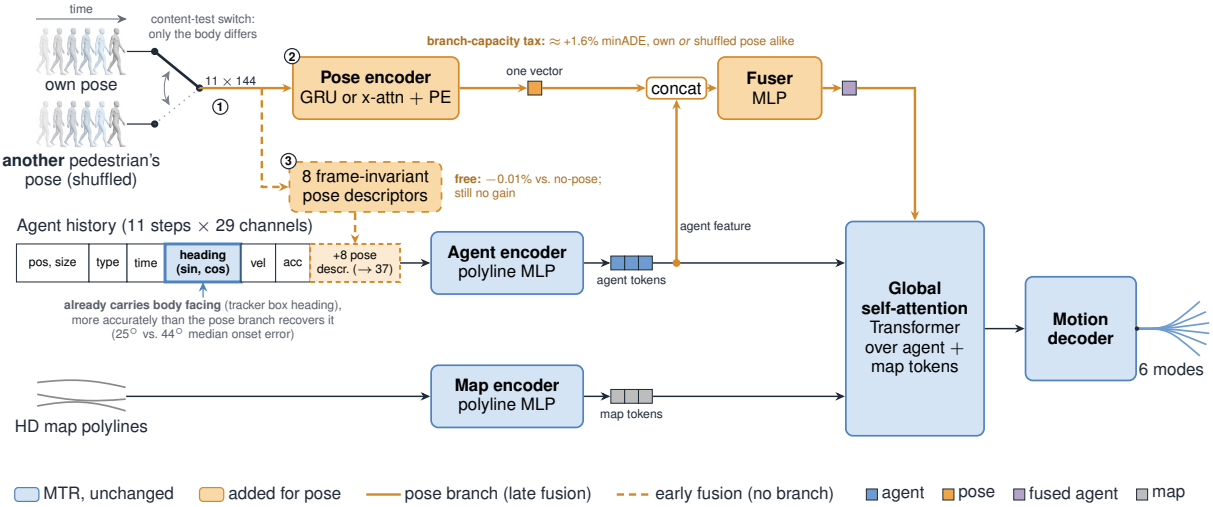


Fig. 1. The two arms differ only in *whose* body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed; everything downstream is shared. Blue: MTR, unchanged; orange: added for pose. The agent-history strip shows that the tracker's box heading is *already* an input. Solid orange (2): the pose branch (late fusion), which costs ≈1.6% minADE whether it carries the right body or the wrong one; dashed (3): early fusion, which removes that cost and still yields no benefit—and which, fed an uncorrupted oracle, converts it on only one of three seeds (Sec. V-C).

A fourth arm, *oracle*, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched; it shows the pathway can carry information at all (Sec. IV-D).

Pose versus no-pose confounds information with branch capacity; pose versus shuffled isolates content. We call the latter the *content test* and treat it as decisive. Relative differences are $(A - B)/B$; negative favours A.

Inference is over *seeds*, which is the unit the treatment is applied to: every comparison between trained arms is a paired t-test on per-seed differences. The two exceptions are flagged where they occur—per-pedestrian subpopulation tests and the per-joint intervals of Table IV, which resample pedestrians. We also report a hierarchical bootstrap that resamples seeds and then pedestrians, and it agrees; we do not quote a percentile bootstrap p-value, because with five seeds whose differences share a sign it returns the resolution floor $1/N_{\text{boot}}$ rather than a calibrated tail probability, and a pedestrian-only bootstrap ignores between-seed variance entirely.

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER RUNG BY A GRID RUN SEPARATELY FOR EVERY ARM; ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS; p-VALUES ARE PAIRED t-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

Training scenes	579	5k	25k
Pedestrians	6,348	31,302	142,807
Seeds	5	3	5
minADE (m)			
No-pose	0.4599	0.4176	0.3716
Pose	0.4620	0.4138	0.3778
Shuffled	0.4589	0.4181	0.3777
Pose vs. no-pose	+0.45%	−0.90%	+1.66%*
Content test	+0.66%	−1.02%	+0.04%
p (content)	0.52	0.56	0.93
minFDE (m)			
No-pose	0.9922	0.8992	0.8049
Pose	0.9824	0.8875	0.8131
Shuffled	0.9780	0.8900	0.8104
Pose vs. no-pose	−0.99%	−1.30%	+1.02%†
Content test	+0.45%	−0.28%	+0.33%
p (content)	0.63	0.83	0.61

*$t(4)$=10.3, p=5×10^{-4}; the shuffled arm pays the same cost, +1.63%, $t(4)$=2.8, p=0.048.
†$t(4)$=6.1, p=0.004.

C. Protocol

Every arm trains for 30 epochs with AdamW (batch 2 per GPU, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25; the decoder predicts K=6 modes. The learning rate is chosen per rung from a grid run separately for each arm: 1×10^{-4} at 579 scenes, 5×10^{-5} at 5k and 2.5×10^{-5} at 25k, where every arm peaks at the same interior optimum. Validation is fixed at 5,171 scenes and 16,486 pedestrians; the directed search of Sec. IV-C splits it once into a development half (2,617 scenes, 8,445 agents) and a held-out half (2,554 scenes, 8,041 agents). Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP and gated miss rate need the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

TABLE II
CLEAN CORE AT 579 SCENES, n=5 SEEDS, BOTH REGIMES. THE SHUFFLED ARM HAS THE LOWEST MEAN IN BOTH: ANOTHER PEDESTRIAN'S BODY IS NOMINALLY THE BEST INPUT.

Arm	Per-seed minADE	Mean ± sd	Content
From scratch			
No-pose	.458 .443 .453 .495 .450	0.4599 ± 0.0206	—
Shuffled	.454 .467 .451 .455 .467	0.4589 ± 0.0075	—
Pose	.471 .456 .453 .457 .474	0.4620 ± 0.0094	$+0.66\%$
Fine-tuned on a full-WOMD backbone			
No-pose	.372 .366 .367 .373 .368	0.3692 ± 0.0031	—
Shuffled	.364 .363 .364 .374 .373	0.3676 ± 0.0055	—
Pose	.364 .372 .371 .374 .373	0.3708 ± 0.0040	$+0.88\%$

IV. RESULTS

A. The null, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%; the content test is null at every rung, in minADE and in minFDE alike, and in miss rate at 2 m, where the content test is $+2.3\%$ at 25k (p=0.21) and $+4.5\%$ fine-tuned (p=0.19)—null, and with its sign against pose on all three metrics. The rungs are not equally sensitive, and we state what each one could have seen: at 80% power the minimum detectable content effect is 3.5% at 579 scenes (n=5), 7.6% at 5k (n=3) and 2.2% at 25k (n=5). The 5k rung is therefore weak evidence on its own and we do not count it as an independent null; the weight falls on 25k and on the eight-seed confirmation of Sec. IV-C. At 25k the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$, so at the largest scale we train the data exclude a content effect beyond roughly 1.7% in either direction—a bound tighter than the 1.6% cost of the branch that would carry it. The pedestrian-resampling bootstrap gives a visibly narrower interval ($[-0.0040, +0.0047]$ m, $\approx \pm 1.2\%$); with five clusters it undercovers, and we quote the wider seed-level bound throughout.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning (n=5), the content test is $+0.88\%$ (p=0.18), again with its sign against pose (Table II).

Seed count decides what Table II appears to say. On the first three seeds the same runs read as *pose harms* from scratch ($+1.85\%$) and *shuffled is best* after fine-tuning (-1.28%); at n=5 both shrink roughly fourfold. Both three-seed effects are significant under a pedestrian-resampling bootstrap (p=0.037, p=0.016) and neither is under a paired t over the same seeds (p=0.18, p=0.12): three seeds do not resolve a 1% effect here, and a bootstrap over the wrong unit will not say so.

B. Two effects that are real, and are not pose

A branch-capacity tax. At 25k both pose arms are worse than no-pose ($+1.66\%$, $p=5\times10^{-4}$; $+1.63\%$, p=0.048) and indistinguishable from each other. An extra input branch costs $\sim 1.6\%$ whether it carries the right body or the wrong one.

This is an operating-point statement, and the operating point matters: it is measured at each run's best validation epoch. Models with wider inputs overfit sooner, so the sign of a pose-versus-no-pose comparison is not stable under a fixed late-epoch rule, where the no-pose arm has degraded further by then ($+8.0\%$ in favour of pose at epoch 30, n=3; $+0.9\%$ at n=8, both n.s. and neither resolved at these seed counts). What is stable is that the comparison moves with the stopping rule, so we never compare architecturally different arms at a fixed late epoch. The *content* test—whose two arms have identical input dimensionality, and for which this asymmetry does not arise—is null under either rule. Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels removes the cost (-0.01% versus no-pose, p=0.99; -1.67% versus the branch, p=0.12 at n=3) without creating a benefit: pose can be carried for free, and still does not pay. That freedom turns out to be partly illusory—Sec. V-C shows this pathway fails to exploit even an uncorrupted oracle on two seeds in three, so a channel it costs nothing to carry is also a channel it largely does not read. We quote the seed-level p here for consistency with the rest of the paper; a pedestrian-resampling bootstrap makes the branch contrast look decisive (p=0.006), which is exactly the unit error Sec. III warns against.

Tuning dominates the treatment. At 5k the learning rate inherited from the smaller rung made pose look catastrophic ($+37\%$); with per-arm tuning, -0.90%, n.s. Across the 25k grid the learning rate moves minADE by up to 16.7% against a content signal of $\sim 0.05\%$: a pose-versus-baseline comparison that does not tune each arm separately is a claim about the learning rate.

C. Search and confirmation

Hand-designed integrations sample a small space, so we ran a directed search in which a coding agent rewrote the pose-integration code against a frozen training repository, scoring on a development split and never seeing the held-out split. Round one scored 13 candidates and accepted none. Round two produced one acceptance: a stationary-gated body-facing plus foot-contact descriptor that beat the incumbent on $5/5$ seeds and reached -1.21% content on the held-out split, clearing the pre-set 0.8% acceptance floor—under a truncated 12-epoch proxy recipe.

Because that acceptance rested on a truncated recipe, we pre-registered a confirmation before running it: three arms, eight seeds, 30 epochs, primary endpoint the held-out split at the fixed last epoch. It returned -0.04%, $t(7) = -0.11$, p=0.92. The proxy signal was a convergence-rate artifact—the arms differ in how fast they train, not in where they end—and a truncated proxy cannot settle a sub-1% effect. We report the acceptance rather than only the confirmation because the search behaved honestly: it found the best candidate the representation admits, and pre-registration is what removed it.

D. The branch is not inert

A null measured through one pathway invites the obvious objection that the pathway is dead. The shuffled control does not dispose of it: it shows the model cannot distinguish its own body from a stranger's, which is equally consistent with

a branch that transmits nothing at all. The branch-capacity tax shows the branch perturbs the output, not that it carries signal, and the fusion residual is zero-initialised, so a dead branch and an uninformative one begin at the same place.

We therefore ran the positive control. The *oracle* arm replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s — genuinely predictive by construction — keeping the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds the oracle arm reaches 0.2945 against no-pose's 0.4514 on the same seeds—-34.8%, $t(2) = -81.4$, $p = 1.5 \times 10^{-4}$—and -35.6% against the shuffled arm. The oracle is of course not a usable model, since it reads the future; its only job is to calibrate the pathway. It does so emphatically: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given.

E. Where a benefit could still hide

On the confirmation runs the content test is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, -0.04% at 8 s); null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration; and null for standing-then-moving pedestrians, the subpopulation built to favour it (n=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on an expanded validation split; p=0.80 by a pedestrian-resampling bootstrap, the one place we quote one, because this subpopulation is defined per pedestrian).

Across the roughly forty content tests this study reports, one reaches $p<0.05$: in the stationary subpopulation of the heading ablation, correct pose is $+1.8\%$ *worse* than a stranger's (n=8, p=0.011). At that multiplicity one such result is expected by chance, and its direction is against a pose mechanism rather than for one; we report it so that the released metrics contain no surprise.

V. Why Pose Does Not Pay

A. What the oracle ladder measures

The content test bounds what pose is worth, and the oracle of Sec. IV-D shows the branch can carry information. Neither says what *kind* of information this pathway turns into accuracy. We measured that by degrading the oracle.

Each rung of Table III feeds the branch the same direction-only oracle—the bearing of the agent's own future displacement at 3 and 8 s—corrupted by a fixed angular error, and is labelled by the median bearing error actually delivered: 0.6745σ by construction, since the corruption is a fixed per-agent rotation, and measured to within 0.7° of it on 11,328 scored agents. Displacement magnitude is withheld, so the channel carries direction alone. The error is drawn once per pedestrian, not per epoch: a real estimator's error is a property of the observation, and resampling it would let training average it away.

At 25k scenes every rung converts, and the gain stays large: -34.50% with an exact bearing, -19.77% at 5.0°, -15.68% at 10.0° and -13.57% at 15.1°, all at $p \le 0.0016$ against minimum detectable effects of 1.4–2.9 pp. The decay is not exponential: one exponential fitted to the four rungs leaves residuals (measured minus fitted) of -1.5, $+3.2$, $+0.3$ and -2.5 pp, a systematic pattern in θ against those MDEs rather than scatter about the curve. The curve decelerates, and two terms fit it to 0.4 pp:

$$\Delta minADE(\theta) \;\approx\; -\big[\,13.1\% + 21.4\% \cdot e^{-\theta/4.4^\circ}\,\big], \qquad (1)$$

a precision term with a half-life of 3.0° over a term that survives every corruption we applied. The seeds agree closely: 12.3–14.2% persistent, amplitudes 20.8–21.8%, half-lives 2.8–3.4°.

The angular axis is not an acceptance threshold. It is tempting to read θ as the accuracy a cue must reach—feed the branch a channel whose bearing error is θ, expect the gain at rung θ. The ladder refutes that reading itself. The baseline already receives the tracker's box heading, as a per-timestep sine–cosine pair for every agent (Fig. 1), and that heading estimates the *same* future displacement bearing at 3.9° in median—better than the 15.1° rung's channel at every percentile (9.3° against 25.8° at the third quartile, 20.1° against 36.8° at the ninetieth). A channel uniformly less accurate than an input the model already has still buys -13.6%. Nor does the branch pay for merely being occupied: an extra branch carrying uninformative content *costs* $\sim$1.6% (Sec. IV-B), and the same branch fed another pedestrian's pose returns $+0.04\%$ (Table I). The 34.5% is content, and the only content this channel has is the realised future.

The surviving term is future-side mode selection. Eq. (1) splits the gain, and the two terms behave differently in ways that identify them. Per-agent records at each run's own best epoch, paired between arms on 16,486 agents and averaged over seeds, put the persistent term at the end of the trajectory: refitting the decomposition at each horizon leaves the precision half-life at 2.9–3.9° throughout, while the persistent share of the gain rises from 21% of the 2 s error reduction to 39% at 8 s and 53% at the endpoint. The selection reading predicts exactly that, and it predicts more: corrupting the bearing should barely affect *which* of the six modes is right. It does not. The gap between the top-ranked mode and the best of six—what the forecaster loses by choosing wrongly—falls from 0.67 m to 0.26 m with an exact oracle, and still to 0.29 m at 15.1°: the corrupted channel closes 92% of the perfect oracle's selection gap while recovering 39% of its minADE. The absolute gain is also largest in the quartile where the baseline's own modes disagree most at the endpoint (0.150 m against 0.111–0.130 m elsewhere, at θ=0). A coarse future bearing says which way the agent went; only a precise one refines where along that way, and that refinement is what 3° of error destroys.

The ladder is therefore a positive control and an upper bound—what any direction channel through this branch could

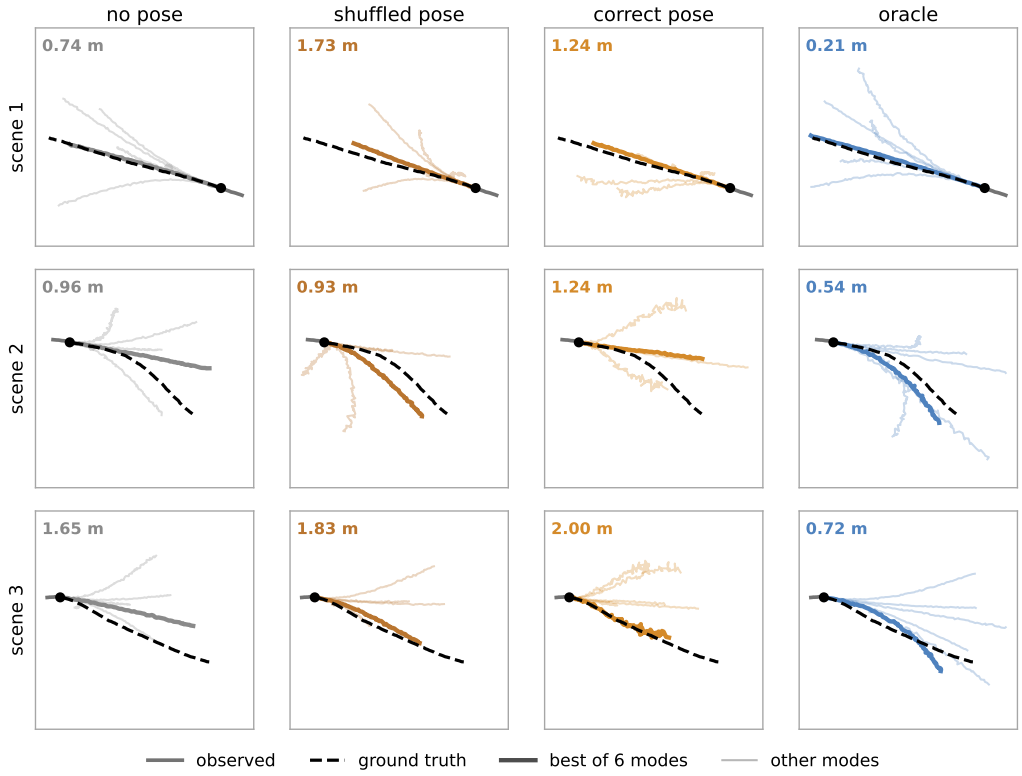


Fig. 2. Qualitative comparison of the four arms on the same three agents (seed 101, epoch 30, the epoch for which full trajectory dumps were retained). Scenes are picked by percentile of the *no-pose* arm's minADE (60th, 76th, 92nd) among agents with a full 8 s future and more than 3 m of travel—selection is on the baseline alone, never on the treatment. Thin lines are the six predicted modes, thick the best of them, dashed the ground truth, grey the observed past. Fed the agent's own future bearing, the oracle bends its modes onto the true path through the *same* branch that carries pose; correct pose is not consistently distinguishable from a stranger's pose or from no pose. Per-panel values are single-seed illustrations: across seeds the oracle's advantage is stable (-34 to -40%), whereas the content difference at this fixed epoch is not (-16.2, -8.0, -0.2% for seeds 101/202/303) and is null at the best-epoch operating point used for inference (Table I).

buy *if it knew the future*—and (1) says how much of that bound is precision rather than selection. Selection is not available to any observation-time cue at any accuracy, because it is knowledge of which future occurred.

A null at low power looks like a threshold. We first ran this ladder at 579 scenes, where it reads as a clean detection floor: significant at $10.0°$ (-6.0%, p=0.04), indistinguishable from zero at $15.1°$ (-1.4%, p=0.47), flat above. At 25k that second rung is -13.6%. The 579 ladder could never have seen it: scaling its own θ=0 amplitude (-15.8%) by the 25k decay predicts $\approx$6.2% there, against that rung's MDE of 6.7 pp. Its apparent floor was the angle at which a three-seed experiment went blind, and a threshold read off the first non-significant rung is a statement about power wearing the clothes of a property of the task. We report both ladders in Table III for that reason.

B. Why body facing does not pay

The oracle converts a bearing read off the realised future. Body facing is not that: it is an observation-time estimate of a quantity the model already measures. Three properties of it, on the population the ladder scores, say what is left for it to contribute.

It is a noisier copy of an existing input. The box heading estimates the future displacement bearing at $3.9°$ (+3 s) and $7.1°$ (+8 s) in median; SMPL facing reaches $12.3°$ (n=5,824) and $15.1°$ (n=6,206). Box heading is the better estimate at every percentile we measured, the two differ by $10.0°$ in per-agent median, and both derive from the same returns—indeed we calibrated the SMPL forward axis *against* the box heading before measuring either.

Most of its accuracy is the trajectory restated. 95.5% of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than $90°$ (Sec. III). Movers sit at $11.9°$; the 262 stationary agents—the only ones whose facing their velocity does not already imply—sit at $41.2°$. The pooled median largely measures the snapping, and the distribution is heavy either way: $5.5°$ at the first quartile, $23.9°$ at the third, $45.9°$ at the ninetieth percentile.

Where it would be non-redundant, it is worst. On the one case where velocity carries no directional information—which way a stationary pedestrian steps off—SMPL facing predicts onset direction at roughly twice chance over 19,939 validation pedestrians: median error $44.2°$, 50.6% within $45°$ (chance

TABLE III

THE ORACLE LADDER. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH RUNG LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED; PAIRED t AGAINST EACH SEED'S OWN NO-POSE RUN, NEGATIVE FAVOURS THE ORACLE. MDE IS THE MINIMUM EFFECT A RUNG COULD DETECT AT 80% POWER GIVEN ITS REALISED SPREAD. AT 25K SCENES EVERY RUNG CONVERTS—INCLUDING 15.1°, WHICH IS WORSE AT EVERY PERCENTILE THAN THE BOX HEADING THE MODEL ALREADY RECEIVES—SO THE LADDER BOUNDS WHAT A DIRECTION CHANNEL COULD BUY RATHER THAN SETTING AN ACCURACY A CUE MUST REACH (SEC. V-A). THE SAME LADDER AT 579 SCENES REPORTS A FLOOR AT 15.1° THAT IS ITS OWN MDE: PROPORTIONALLY TO ITS θ=0 AMPLITUDE IT SHOULD HAVE READ ≈6.2% THERE, AND IT COULD RESOLVE 6.7.

Bearing error	n	ΔminADE	p	95% CI	MDE
25k scenes					
0.0°	3	−34.50%	0.0001	[−35.6, −33.4]	1.4
5.0°	3	−19.77%	0.0007	[−22.1, −17.5]	2.9
10.0°	3	−15.68%	0.0009	[−17.7, −13.6]	2.6
15.1°	3	−13.57%	0.0016	[−15.9, −11.2]	2.9
579 scenes: the same ladder, underpowered					
0.0°	3	−15.8%	0.021	[−25.6, −5.9]	12.3
5.0°	3	−7.6%	0.0004	[−8.2, −6.9]	0.8
10.0°	5	−6.0%	0.040	[−11.6, −0.5]	7.4
15.1°	5	−1.4%	0.472	[−6.5, +3.6]	6.7
24.9–60.1°	3	−0.1 to +2.7%	≥ 0.24	—	5.5–9.1

90°, 25%). Informative, consistent with the 41.2° above, and not an artifact of our axis fitting: the box heading estimates the same quantity at 25.2°, and paired on the same events SMPL is 7.3° worse in median (Wilcoxon p=2.6×10^{-43}). Nor does pose take over where perception degrades—over 1,705 onset events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and *widens* with range, to +32.9° at 41–75 m, because both derive from the same returns.

Facing is therefore redundant where it is accurate and inaccurate where it would be non-redundant, and neither half of that squeeze is the selection signal the oracle trades on. What survives it is what the content test measures directly: at 25k scenes +0.04%, with the data bounding it to ±1.7%—inside the ∼1.6% cost of the branch that would carry it.

What this does not establish. Redundancy says why little is left for facing to add; it is not by itself a mechanism that blocks conversion, and we tested the implied counterfactual directly. Removing the heading channel entirely and repeating the content test changes it by −0.6 pp in the stationary subpopulation (n=8, p=0.65, CI [−3.5, +2.3]): no effect. That is what the second and third properties above predict—facing is the worse estimate *there* too, at 41.2° against the heading's 25.2°, so deleting its competitor cannot rescue it—but the corresponding test on movers is unrun, and with it the possibility that facing would pay for a forecaster with no heading input. The reading we cannot exclude, and which the content test's own bound asserts, is that pose carries nothing this forecaster can use at any level of redundancy. At n=3 this same comparison read −4.1 pp with all three seeds agreeing in sign—the third multi-percent effect here to dissolve under more seeds.

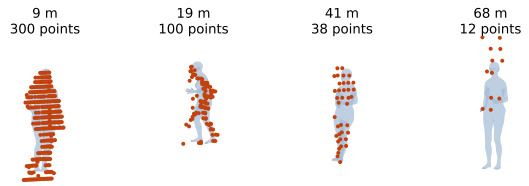


Fig. 3. What the pose estimator has to work with. Raw per-pedestrian LiDAR returns (orange) and the SMPL mesh fitted to them (blue), at four ranges and a common metric scale. Each panel is a frame whose return count equals the median for its range band over the validation segments: 300, 100, 38 and 12 points. Past ≈40 m a pedestrian is a few dozen returns, mostly on head and torso, while the mesh over them still has fully articulated limbs: at that range the articulation is largely the AMASS prior. That applies to the far panels only—scored against ground-truth keypoints across all ranges, every limb joint carries real skill, and it is the *head* the pipeline emits from its prior (Table IV). Sparsity at range is therefore one contributor to the facing error of Sec. V-B, where the pose-versus-heading gap widens to +32.9° at 41–75 m, not its whole explanation.

C. The cheaper pathway does not rescue it

Early fusion removes the branch tax (Sec. IV-B), which makes it the pathway worth recommending—so we ran the same ladder through it, carrying the oracle in the per-timestep agent channels instead of the branch. At a given rung the six channels equal the branch's six inputs bit for bit, so the rungs keep their measured-error labels.

No rung is significant here, *including the uncorrupted one*: −4.5% at θ=0 (p=0.44), −1.0% at 5.0° (p=0.53) and +0.2% at 10.0° (p=0.94), at n=3 throughout. We do not read a floor, or its absence, into that: the MDE at θ=0 is 25.5 pp, and Sec. V-A is a worked example of what an MDE that size does to a ladder. At this power the significance column cannot separate a pathway that ignores the channel from one that converts it.

The per-seed pattern can, and it is the evidence we rest on. At θ=0 the channel is the agent's own future bearing; the side branch converts it on all three seeds (−16.5%, −19.3%, −11.5%) while early fusion converts it on one (−14.0%, against +0.7% and −0.3%), and the two failing seeds sit at their no-pose level from θ=0 onward rather than scattering about it. A difference in *how many seeds move at all* is not something a wide interval explains away.

This bears on the free-carriage result of Sec. IV-B. "No tax, no benefit" admits two readings: that early fusion integrates pose at no capacity cost, or that it largely ignores the appended channels, in which case there is nothing to cost and nothing to gain. A perfect channel separates them—the first predicts a large gain, the second none—and two seeds in three show none. We therefore do not offer early fusion as a rescue for a cue that fails through the branch. We stop short of the stronger claim that it cannot convert one: that needs this ladder at 25k, where the branch ladder's MDE was 1.4–2.9 pp at the same n=3. Caveats: n=3, and the free-carriage result was measured at 25k scenes while this ladder runs at 579.

TABLE IV

PER-JOINT RECOVERABILITY AGAINST GROUND-TRUTH KEYPOINTS. MP IS THE ERROR OF ALWAYS PREDICTING THE MEAN POSE (EQUAL TO THE JOINT'S GROUND-TRUTH SPREAD BY CONSTRUCTION); $\mathrm{SKILL} = 1 - \mathrm{MPJPE}/\mathrm{MP}$. THE HEAD'S MPJPE BEATS BOTH WRISTS WHILE CARRYING A TENTH OF THEIR SKILL. INTERVALS RESAMPLE PEDESTRIANS (n=534), NOT FRAMES, SINCE FRAMES WITHIN A TRACK ARE STRONGLY CORRELATED. EVERY LIMB INTERVAL CLEARS ZERO BY A WIDE MARGIN (LOWEST BOUND $+0.43$); THE HEAD'S IS $[-0.003, +0.089]$, THE ONE INTERVAL THAT CONTAINS IT.

Joint	MPJPE (mm)	MP (mm)	SKILL
Elbows	91.2 / 91.5	252.9 / 248.9	+0.64 / +0.63
Hips	39.8	97.9	+0.59
Wrists	118.3 / 125.4	271.1 / 276.4	+0.56 / +0.55
Shoulders	87.3 / 89.0	171.5 / 164.0	+0.49 / +0.46
Ankles	114.0 / 116.6	221.6 / 223.8	+0.49 / +0.48
Knees	78.4 / 81.5	150.8 / 151.2	+0.48 / +0.46
Head	**111.6**	**116.9**	**+0.045**

D. The complementary cue is barely measured

Gaze precedes locomotion, so head yaw relative to the torso is a plausible *independent* cue—and it is very nearly not recoverable here. In the fitted meshes head-versus-torso yaw has a standard deviation of $3.8°$, while against the ground-truth 3D keypoints released with Waymo Perception [16], [17] (n=8,234) the real quantity varies $19.2°$: a fivefold under-dispersion, so the estimator is largely emitting its prior. Consistently, head yaw carries no usable signal for onset direction ($r = -0.03$, n.s.), and the head is the one joint in Table IV whose interval includes zero. At 20–40 m the head is a near-spherical handful of points with little geometry to constrain yaw.

E. MPJPE is the wrong acceptance metric

A single mean error cannot distinguish an estimator that tracks a joint imprecisely from one that emits its prior. We therefore score each joint against the strongest trivial competitor, always predicting the mean root-aligned pose: $\mathrm{SKILL}_j = 1 - \mathrm{MPJPE}_j/\mathrm{MP}_j$, where values at or below zero mean the estimator contributes nothing at that joint.

Evaluated on Waymo Perception against ground-truth keypoints (1,789 frames, 534 pedestrians), every limb joint scores $+0.457$ to $+0.639$ (Table IV). The head scores $+0.045$—yet its MPJPE (111.6 mm) is *better* than either wrist (118.3/125.4 mm). Its error is flattering not because it is well estimated but because it barely improves on its own mean-pose baseline (116.9 mm), whereas the hips—whose ground-truth spread is *smaller* at 97.9 mm—are predicted at 39.8 mm. Across the thirteen joints MPJPE and skill are essentially uncorrelated (Spearman $\rho = -0.11$, p=0.72): the aggregate is not merely a noisy proxy for information content but carries almost none of it, and for the head it actively inverts the ordering. We recommend reporting per-joint skill alongside it whenever a pose estimator is accepted into a downstream stack.

F. A general condition

Two results locate where pose could have helped. At 2 s, minADE is 0.091 m—about 3% of the displacement—so little error is left for an extra input to remove. This is a property of the forecaster's own residual, not of constant velocity: single-mode constant-velocity extrapolation [18] gives 0.310 m at 2 s on the same set, so short-horizon forecasting is *not* CV-saturated (a comparison that favours the model, since minADE takes the best of K=6 modes). Saturation here means the baseline's own remaining error is a few percent of the displacement. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose; stratifying by how badly constant velocity fails still yields no effect.

In general form: *an auxiliary modality improves a forecast only where the baseline is neither saturated nor dominated by intent the modality cannot observe, and where the modality measures the cue it is meant to carry more accurately than the inputs the model already receives.* All three clauses must hold. The third is the cheapest to check—it is a measurement on the dataset, with no training at all—and the one most often skipped. For pose on WOMD it is the one that fails: the box heading carries the same bearing three times more accurately, and on the stationary pedestrians where it does not, facing is worse still.

VI. WHAT THIS MEANS IN PRACTICE

For an AV stack, four consequences follow. First, before integrating an auxiliary modality, measure its error on the cue it is meant to carry *against the inputs the model already receives*. That costs no training and it is what decides pose here. Second, an oracle ladder is still worth its handful of runs, as a positive control and as an upper bound on the pathway—but size it so its MDE is below the effect you would act on, and do not read its angular axis as an acceptance threshold: ours converts a channel less accurate than an input the model already has, because an oracle knows which future occurred and no sensor does (Sec. V-A). Third, the capacity cost of an extra branch is real ($\sim$1.6% here, at the best-epoch operating point) and is easily mistaken for a modality effect in either direction; early fusion avoids it, though on our evidence it does so partly by not reading the channel, and a shuffled-pose control separates the two. Fourth, do not accept a pose estimator on aggregate MPJPE alone: check per-joint skill, because the joints an aggregate flatters may be the ones carrying a prior.

Limitations. The largest rung we train is 25k scenes of the 299,220 exported, so the full rung is a compute limit rather than an annotation one; at eight seeds the minimum detectable content effect is $\approx$1.25% at 80% power, and the ladders run at n=3 per rung (Table III). We test one forecaster, one dataset and one pose estimator. We ladder two integration pathways rather than one, but two is not every way a cue can enter a model, and only the late-fusion ladder is resolved at 25k—the early-fusion one remains at 579 scenes, where this paper's own power argument applies to it (Sec. V-C). The redundancy

counterfactual—deleting the heading channel to see whether facing then pays—is run only on the stationary subpopulation, where facing is also the worse estimate, so we cannot say what a forecaster without a heading input would do with pose on movers. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale; on Waymo Perception, where raw imagery exists, an RGB estimator (HMR2.0) reaches 106.2 mm root-aligned MPJPE against LiDAR-HMR's 87.6 mm on the same 347 frames, so the obvious better-pose route is the worse one here. We do not model pose–pose interaction.

VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD. The null is controlled against a shuffled-pose baseline and holds across scale, representation, integration, a directed search and a pre-registered confirmation; an oracle through the same branch rules out a dead pathway. It also has a quantitative mechanism. An oracle ladder shows this integration path converts a direction channel at every accuracy we trained, including one uniformly worse than the tracker heading the model already receives, so the pathway is not the obstacle and no attainable pose accuracy is the remedy. What the oracle sells is knowledge of which future occurred: half its endpoint gain is mode selection that survives arbitrary angular corruption, against a fifth at 2 s, and a channel corrupted to 15.1° still closes 92% of the selection gap. Body facing cannot supply that at any accuracy, and as a bearing it is a noisier copy of one the model already has—worst, at 41.2°, on the stationary pedestrians the cue was introduced to serve. Fusing the same cue early removes the branch's capacity cost but does not rescue it: that pathway fails to exploit even an uncorrupted bearing on two seeds in three, so both integration routes fail, for different and identifiable reasons. The lesson generalises past pose—measure a modality's error on the cue it is meant to carry against the inputs the model already receives, before integrating it, and make sure any ladder you judge it against can resolve the effect you would act on.

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